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STALLING CHARACTERISTICS OF THE SUPERMARINE SPITFIRE VA AIRPLANE

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WASHINGTON

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desirable in some respects than those of any of the pursuit-type airplanes formerly tested in a similar manner. This paper was originally issued as a memorandum report to the Materiel Division, Army Air Forces.

APPARATUS AND TESTS

The Supermarine Spitfire is a single-engine, single-place, low-wing cantilever monoplane with retractable landing gear and partial-span split flaps. (See figs. 1, 2, 3, and 4.)

Throughout these tests of stalling characteristics the gross weight of the airplane was maintained at approximately 6184 pounds and the center of gravity was at approximately 31.4 percent of the mean aerodynamic chord.

NACA recording instruments were used to record the following items:

1. Airspeed.
2. Positions of the rudder, elevators, and ailerons.
3. Angular velocity in roll.
4. Angle of sideslip.
5. Normal, longitudinal, and transverse accelerations.
6. Elevator force or rudder force (in some cases only).

These instruments recorded photographically and were synchronized by means of a timer.

The airspeed recorder in the Spitfire was calibrated by flying in formation with another airplane. The airspeed recorder in the other airplane was calibrated by means of a trailing airspeed head. The calibration of the pilot's airspeed meter is shown in figure 5.

For most of the stall measurements, the angle of sideslip was measured by means of a yaw vane attached to a boom extending a chord length ahead of the left wing tip. In order to determine whether any error was introduced by an-

With gun ports open, the flow separated from the outboard wing panels before the center section stalled. Figure 16 shows that control was maintained in this condition for several seconds after the stall, but a mild right roll eventually occurred.

MAXIMUM LIFT COEFFICIENTS FOR VARIOUS FLIGHT CONDITIONS

The maximum lift coefficients as measured in various conditions of flight are shown in the accompanying table.

Condition	Flap position	Gear position	Manifold pressure (in. Hg)	Engine speed (rpm)	Gun ports covered		Gun ports open	
					Indicated stalling speed (mph)	Maximum lift coefficient	Indicated stalling speed (mph)	Maximum lift coefficient
Gliding	up	up	throttle closed	----	90	1.18	91	1.15
Cruising	up	up	36.9 (3¾ lb/sq in. boost)	2650	77	1.63	76	1.68
Landing	down	down	throttle closed	----	80	1.51	81	1.46
Landing approach	down	down	21.4 (-4 lb/sq in. boost)	2300	69	1.99	75	1.72
Take-off	up	down	43.9 (7 lb/sq in. boost)	2850	80	1.49	--	----
Climbing	up	up	43.9 (7 lb/sq in. boost)	2850	76	1.64	--	----

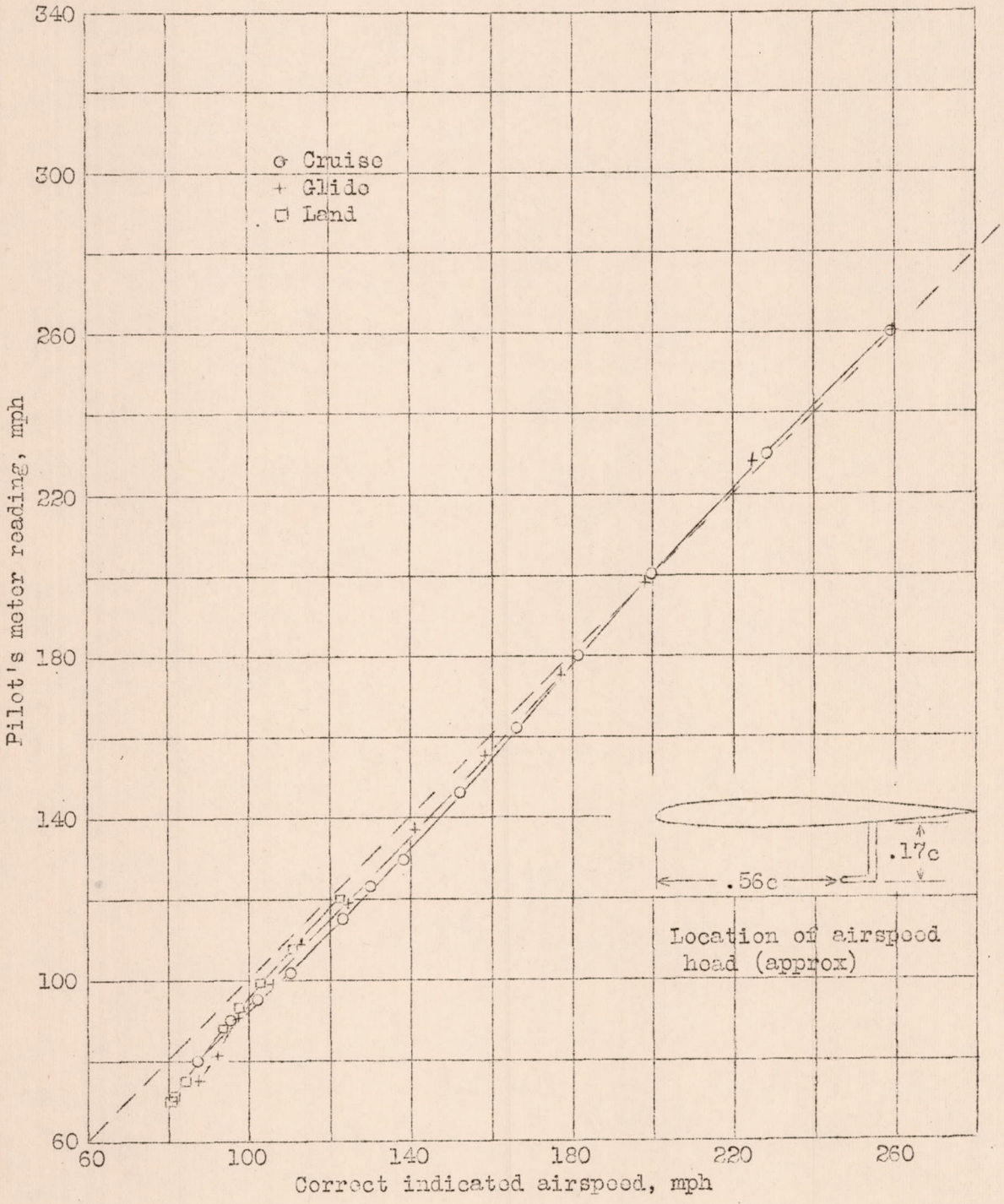


Figure 5.- Calibration of the pilot's airspeed indicator in the Supermarine Spitfire airplane.